

Spin-Ordering in a Planar Classical Heisenberg Model

FRANZ WEGNER

Max-Planck-Institut für Physik und Astrophysik, München, Germany

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We consider a D -dimensional system of classical spins rotating in a plane and interacting via a Heisenberg coupling. The spin-correlation function $g_D(r)$ is calculated for large distances r in a low-temperature approximation (taking into account short-range order):

$$\begin{aligned} g_1(r) &= \exp(-C_1 Tr), \\ g_2(r) &\sim r^{-C_2 T}, \\ \lim_{r \rightarrow \infty} g_3(r) &= \exp(-C_3 T). \end{aligned}$$

In two dimensions this model exhibits infinite susceptibility

$$\chi = \frac{1}{2T} \sum_r g_2(r)$$

at low temperatures.

Comparison is made of g_1 with the exact result and of g_3 with a spinwave-treatment showing agreement of $\ln g(r)$ within order T . Spontaneous magnetization for $D=1$ and 2 is ruled out exactly.

Introduction

Recently MERMIN and WAGNER¹ proved rigorously the absence of spontaneous magnetization in two-dimensional isotropic Heisenberg models with short-range interaction. The proof included the absence of magnetization in the xz -plane for anisotropic interaction $I_x = I_z \neq I_y$. They pointed out that this proof does not rule out another kind of phase transition. KAPLAN and STANLEY² calculated the high-temperature expansion of the susceptibility χ and using Padé-technique they conjectured a behaviour $\chi \sim (T - T_c)^{-\gamma}$ and therefore a phase transition at $T_c > 0$.

Here a system is considered which exhibits within a low-temperature approximation zero spontaneous magnetization and infinite susceptibility in two dimensions.

¹ MERMIN, N. D., and H. WAGNER: Phys. Rev. Letters **17**, 1133 (1966).

² STANLEY, H. E., and T. A. KAPLAN: Phys. Rev. Letters **17**, 913 (1966). — KAPLAN, T. A., and H. E. STANLEY: Bull. Am. Phys. Soc. **12**, 334 (1967).

The Model. Formula for Spin-Correlation Function

The system consists of N classical spins $\mathbf{m}_r$ located at the sites $\mathbf{r}$ of a Bravais lattice, every spin rotating in the xz -plane with one degree of freedom φ_r :

$$\mathbf{m}_r = \mathbf{e}_z \cos \varphi_r + \mathbf{e}_x \sin \varphi_r \quad (1)$$

interacting via

$$H = -\frac{1}{2} \sum_{\mathbf{r} \mathbf{r}'} I(\mathbf{r} - \mathbf{r}') \mathbf{m}_r \mathbf{m}_{r'} = -\frac{1}{2} \sum_{\mathbf{r} \mathbf{r}'} I(\mathbf{r} - \mathbf{r}') \cos(\varphi_r - \varphi_{r'}). \quad (2)$$

The interaction is assumed to be short-range and attractive. Then at low temperatures short-range order of the spins appears. Therefore $\cos(\varphi_r - \varphi_{r'})$ is expanded in powers of $\Delta\varphi$ in the Hamiltonian and terms of order $\Delta\varphi^4$ are neglected:

$$H = -\frac{N}{2} \sum_{\mathbf{r}} I(\mathbf{r}) + \frac{1}{4} \sum_{\mathbf{r} \mathbf{r}'} I(\mathbf{r} - \mathbf{r}') (\varphi_r - \varphi_{r'})^2. \quad (3)$$

This model can be regarded as a system of infinite ($S = \infty$) Heisenberg spins with anisotropic interaction $I_x = I_z > |I_y|$. Then apart from small oscillations in y -direction the spins rotate in the xz -plane at temperatures $T \ll (I_z - |I_y|)/k_B$. The correlation function according to (3) is calculated exactly. By introducing the Fourier-transformed

$$\varphi_{\mathbf{k}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}} e^{-i\mathbf{k}\mathbf{r}} \varphi_r, \quad \varphi_r = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k}\mathbf{r}} \varphi_{\mathbf{k}} \quad (4)$$

and

$$\varepsilon_{\mathbf{k}} = \sum_{\mathbf{r}} I(\mathbf{r}) (1 - \cos \mathbf{k}\mathbf{r}) = 2 \sum_{\mathbf{r}} I(\mathbf{r}) \sin^2 \left(\frac{\mathbf{k}\mathbf{r}}{2} \right) \quad (5)$$

we get

$$H = -\frac{N}{2} \sum_{\mathbf{r}} I(\mathbf{r}) + \frac{1}{2} \sum_{\mathbf{k}} \varepsilon_{\mathbf{k}} \varphi_{\mathbf{k}} \varphi_{-\mathbf{k}} \quad (6)$$

where $\mathbf{k}$ is restricted to the first Brillouin zone. The amplitudes $\varphi_{\mathbf{k}}$ are complex and $\varphi_{-\mathbf{k}} = \varphi_{\mathbf{k}}^*$. Therefore we divide the first Brillouin zone into two halves B_+ and B_- in such a way that one vector of every pair $\mathbf{k}, -\mathbf{k}$ lies in B_+ and the other in B_- . (Those $\mathbf{k}$ with $\varphi_{\mathbf{k}} \equiv \varphi_{-\mathbf{k}}$ should be considered separately, but such a consideration does not change the result for $N \rightarrow \infty$ at all.) We define the real amplitudes $\Psi_{\mathbf{k}}$:

$$\Psi_{\mathbf{k}} = \frac{1}{\sqrt{2}} (\varphi_{\mathbf{k}} + \varphi_{-\mathbf{k}}), \quad \Psi_{-\mathbf{k}} = \frac{1}{i\sqrt{2}} (\varphi_{\mathbf{k}} - \varphi_{-\mathbf{k}}) \quad \text{for } \mathbf{k} \in B_+ \quad (7)$$

and get

$$H = -\frac{N}{2} \sum_{\mathbf{r}} I(\mathbf{r}) + \frac{1}{2} \sum_{\mathbf{k}} \varepsilon_{\mathbf{k}} \Psi_{\mathbf{k}}^2. \quad (8)$$

The spin-correlation function is given by

$$g(\mathbf{r}) = \langle \mathbf{m}_o \mathbf{m}_r \rangle = \langle \cos(\varphi_o - \varphi_r) \rangle = \text{Re} \langle \exp i(\varphi_o - \varphi_r) \rangle \quad (9)$$

with

$$\varphi_o - \varphi_r = \sqrt{\frac{2}{N}} \sum_{\mathbf{k} \in B_+} (\Psi_{\mathbf{k}} (1 - \cos \mathbf{k} \mathbf{r}) + \Psi_{-\mathbf{k}} \sin \mathbf{k} \mathbf{r}). \quad (10)$$

From these equations we get

$$\begin{aligned} g(\mathbf{r}) &= \text{Re} \frac{\int d^N \Psi_{\mathbf{k}} \exp(-\beta H + i(\varphi_o - \varphi_r))}{\int d^N \Psi_{\mathbf{k}} \exp(-\beta H)} \\ &= \exp \left(-\frac{2k_B T}{N} \sum_{\mathbf{k}} \frac{\sin^2 \left(\frac{\mathbf{k} \mathbf{r}}{2} \right)}{\varepsilon_{\mathbf{k}}} \right) \end{aligned} \quad (11)$$

with $\beta = 1/k_B T$ (k_B Boltzmann constant).

Evaluation of the Spin-Correlation Function for Nearest Neighbour Interaction

On account of simplicity we regard a primitive linear resp. orthogonal resp. cubic lattice with lattice constant unity and nearest neighbour interaction

$$I(\mathbf{r}) = \begin{cases} I & \text{for nearest neighbours} \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$\varepsilon_{\mathbf{k}} = 4I \sum_{i=1}^D \sin^2 \frac{k_i}{2} \approx I k^2 \quad \text{for small } k. \quad (12)$$

In the thermodynamic limit $N \rightarrow \infty$:

$$g_D(\mathbf{r}) = \exp \left(-\frac{k_B T}{I} f_D(\mathbf{r}) \right), \quad f_D(\mathbf{r}) = \frac{2}{\pi^D} \int_0^\pi d^D k \frac{\sin^2 \left(\frac{\mathbf{k} \mathbf{r}}{2} \right)}{\varepsilon_{\mathbf{k}}/I}. \quad (13)$$

We split f_D into two parts $f_D = f_{\text{reg}} + f_{\text{sing}}$

$$\begin{aligned} f_{\text{reg}}(\mathbf{r}) &= \frac{2}{\pi^D} \int_0^\pi d^D k \sin^2 \left(\frac{\mathbf{k} \mathbf{r}}{2} \right) \left(\frac{I}{\varepsilon_{\mathbf{k}}} - \frac{1}{k^2} \right) \\ f_{\text{sing}}(\mathbf{r}) &= \frac{2}{\pi^D} \int_0^\pi d^D k \sin^2 \left(\frac{\mathbf{k} \mathbf{r}}{2} \right) / k^2. \end{aligned} \quad (14)$$

f_{reg} shows a regular behaviour for $r \rightarrow \infty$, since $I/\varepsilon_{\mathbf{k}} - 1/k^2$ is bound. We replace $\sin^2$ by its mean value $\frac{1}{2}$.

$$f_{\text{reg}}(\infty) = \pi^{-D} \int_0^\pi d^D k (I/\varepsilon_{\mathbf{k}} - 1/k^2) \quad (15)$$

f_{sing} gives the essential r -dependence of the long-range behaviour. We substitute

$$x = \frac{k|r|}{2}$$

and define $e = r/|r|$, then

$$f_{\text{sing}}(r) = \frac{2}{\pi^D} \left(\frac{r}{2}\right)^{2-D} \int_0^{\pi r/2} d^D x \frac{\sin^2(xe)}{x^2}. \quad (16)$$

For $D=1$ the integral in (16) converges. For large r

$$f_{\text{sing}}(r) \approx \frac{r}{\pi} \int_0^\infty dx \frac{\sin^2 x}{x^2} - \frac{r}{\pi} \int_{\pi r/2}^\infty dx \frac{1}{2x^2} = \frac{r}{2} - \frac{1}{\pi^2}. \quad (17)$$

With $f_{\text{reg}}(\infty) = 1/\pi^2$ we get

$$g_1(r) = \exp\left(-\frac{k_B T|r|}{2I}\right). \quad (18)$$

In two dimensions the integral (16) diverges logarithmically for large r :

$$\begin{aligned} \frac{\partial f_{\text{sing}}(r)}{\partial r} &= \frac{1}{\pi} \int_0^{\pi r/2} dx_1 \frac{\sin^2\left(x_1 e_1 + \frac{\pi r}{2} e_2\right)}{x_1^2 + \left(\frac{\pi r}{2}\right)^2} \\ &+ \frac{1}{\pi} \int_0^{\pi r/2} dx_2 \frac{\sin^2\left(x_2 e_2 + \frac{\pi r}{2} e_1\right)}{x_2^2 + \left(\frac{\pi r}{2}\right)^2} \end{aligned} \quad (19)$$

Here as in the second integral of (17) we replace $\sin^2$ by $\frac{1}{2}$.

$$\frac{\partial}{\partial r} f_{\text{sing}}(r) = \frac{1}{2r\pi}, \quad f_{\text{sing}}(r) = \text{const} + \frac{1}{2\pi} \ln r, \quad (20)$$

$$g_2(r) = \exp\left(-\frac{k_B T}{I} \left(f_{\text{reg}} + \text{const} + \frac{1}{2\pi} \ln r\right)\right) \sim r^{-k_B T/2\pi I}. \quad (21)$$

For $D=3$ we take the limit $r \rightarrow \infty$ in (13) by setting $\sin^2 = \frac{1}{2}$. The integral is converging for small k .

$$\lim_{r \rightarrow \infty} g_3(r) = \exp(-k_B T f_3(\infty)/I), \quad f_3(\infty) = \frac{1}{\pi^3} \int_0^\pi d^3 k \frac{I}{\varepsilon_k}. \quad (22)$$

Comparison with Results Received by Other Methods

The results of the foregoing section were derived by an approximation. To confirm them they are compared with three results received by other methods:

1. The one-dimensional case can be treated exactly:

$$\exp(-\beta H) = \prod_r \exp(\beta I \cos(\varphi_r - \varphi_{r+1})) = \prod_r \left(\sum_n c_n e^{i n (\varphi_r - \varphi_{r+1})} \right) \quad (23)$$

with

$$c_n = \frac{1}{2\pi} \int_0^{2\pi} \exp(\beta I \cos \varphi - i n \varphi) d\varphi. \quad (24)$$

Then

$$g_1(r) = \frac{\int d^N \varphi \cos(\varphi_0 - \varphi_r) \exp(-\beta H)}{\int d^N \varphi \exp(-\beta H)} = \left(\frac{c_1}{c_0} \right)^{|r|}. \quad (25)$$

Evaluating c_1/c_0 yields

$$c_1/c_0 = \exp(-k_B T/2I) + O(T^2/I^2) \quad (26)$$

therefore $f_1(r) = (r/2) + rO(T/I)$.

2. Starting from the ground-state $\varphi_r = 0$ in the three-dimensional case and regarding the spin-waves in harmonic approximation we get the spontaneous magnetization

$$m = \frac{1}{N} \sum_r \langle \cos \varphi_r \rangle \approx 1 - \frac{1}{2N} \sum_r \langle \varphi_r^2 \rangle = 1 - \frac{1}{2N} \sum_k \langle \Psi_k^2 \rangle. \quad (27)$$

From equipartition theorem we get $\langle \Psi_k^2 \rangle = k_B T / \varepsilon_k$.

Then

$$m = 1 - \frac{k_B T}{2N} \sum_k \frac{1}{\varepsilon_k} = 1 - \frac{k_B T}{2I} f_3(\infty) \quad (28)$$

in accordance with $m = \sqrt{g_3(\infty)}$ from (22) including order T .

3. An exact proof for the absence of spontaneous magnetization for $T > 0$ in this model in one and two dimensions runs analogous to the proof by MERMIN and WAGNER¹ for the Heisenberg-ferromagnet. Starting from the Bogoliubov-inequality

$$\frac{1}{2} \langle \{A, A^+\}_+ \rangle \langle [[C, H], C^+] \rangle \geq k_B T |\langle [C, A] \rangle|^2 \quad (29)$$

one takes the operators A_k and C_k defined by

$$\begin{aligned} A_k |\{\varphi_r\}\rangle &= \left(\sum_r \cos \mathbf{k} \cdot \mathbf{r} \sin \varphi_r \right) |\{\varphi_r + \delta \varphi \cos \mathbf{k} \cdot \mathbf{r}\}\rangle \\ &+ \left(\sum_r \sin \mathbf{k} \cdot \mathbf{r} \sin \varphi_r \right) |\{\varphi_r + \delta \varphi \sin \mathbf{k} \cdot \mathbf{r}\}\rangle \end{aligned} \quad (30)$$

$$C_k |\{\varphi_r\}\rangle = |\{\varphi_r - \delta \varphi \cos \mathbf{k} \cdot \mathbf{r}\}\rangle + |\{\varphi_r - \delta \varphi \sin \mathbf{k} \cdot \mathbf{r}\}\rangle. \quad (31)$$

Here $|\{\varphi_r\}\rangle$ denotes the state given by the set $\{\varphi_r\}$ of the angles. $\delta\varphi$ is an infinitesimal quantity. Then

$$\begin{aligned} N^2 &\geq \sum_{\mathbf{k}} \frac{1}{2} \langle \{A_{\mathbf{k}}, A_{\mathbf{k}}^+\}_+ \rangle \\ &\geq k_B T \sum_{\mathbf{k}} \frac{|\langle [C_{\mathbf{k}}, A_{\mathbf{k}}] \rangle|^2}{\langle [[C_{\mathbf{k}}, H], C_{\mathbf{k}}^+] \rangle} \geq k_B T m^2 N \sum_{\mathbf{k}} \frac{1}{\varepsilon_{\mathbf{k}} + h m}. \end{aligned} \quad (32)$$

Therefore

$$m^2 \leq \frac{1}{\frac{k_B T}{\pi^D} \int d^D k \frac{1}{\varepsilon_{\mathbf{k}} + h m}}. \quad (33)$$

In the limit $h \rightarrow 0$ m vanishes because of the diverging integral in the denominator.

Infinite Susceptibility in Two Dimensions

According to an applied field

$$-h \sum_{\mathbf{r}} m_{\mathbf{r}}^z$$

the susceptibility χ is

$$\chi = \partial m / \partial h = \frac{\beta}{2} \sum_{\mathbf{r}} g(\mathbf{r})$$

for vanishing magnetization. For $D=2$ this sum diverges at low temperatures. This suggests a phase transition from a low-temperature phase with infinite susceptibility to a high-temperature phase with finite susceptibility. The same correlation-function was found for superconductors by RICE³ and for superfluid helium by KANE and KADANOFF⁴.

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³ RICE, T. M.: Phys. Rev. **140**, A 1889 (1965).

⁴ KANE, J. W., and L. P. KADANOFF: Phys. Rev. **155**, 80 (1967).